Is the Platform Part of the Measurement? Validating LLM-Labelled Public Opinion Across Platforms, Languages and Annotator Models in South and Southeast Asian Digital Publics

Research proposal, PhD in Communications and New Media, National University of Singapore (August 2027 intake) · Asad Ikram · asad.ikram53@gmail.com
Proposed supervision: Dr Subhayan Mukerjee (primary) and Dr Zening Duan (co-supervisor)

1. Problem

Computational communication now reads public opinion off social platforms. Sentiment, stance, incivility and narrative frames are labelled by fine-tuned classifiers or, increasingly, by large language models (Gilardi, Alizadeh and Kubli, 2023), and the resulting scores are compared across platforms, languages and time. Each comparison quietly assumes that the instrument measures the same thing everywhere it reads. That assumption is rarely tested, yet the field keeps finding that it matters. Platforms differ in who posts and how they write (Jaidka, Guntuku and Ungar, 2018); the same discourse carries different moral and gendered language on YouTube and TikTok (Chen, Duan and Kim, 2024); and political behaviour around misinformation is studied on Facebook and WhatsApp side by side (Neyazi, Tan and Kuru, 2026). If a "negative" post means something different on WhatsApp than on Facebook, or in Roman Urdu than in English, then a difference between platforms or groups may be an artefact of the instrument rather than a finding about the public. This matters most in South and Southeast Asia, where online publics are multilingual, code-mixed and spread across many platforms.

2. Prior work and my starting point

Work at CNM shows why the platform cannot be ignored. Mukerjee, Yang and Peng (2023) showed how engagement metrics on Facebook shape what news outlets publish, and Ouyang, Mukerjee and Yang (2026) asked whether that relationship carries over to another platform. Duan et al. (2025) showed that the way a construct is operationalised, here moral content, changes what a text-as-data measure captures. Survey research has long had tools for this problem: measurement invariance, differential item functioning (DIF) and score linking (Jacobs and Wallach, 2021, make the same argument for computational systems).

With Dr Ella Haig I recently set out a protocol that brings these tools to social media (Ikram and Haig, 2026, arXiv:2610.16757). It treats each platform as a survey mode: anchor events fixed in advance, a generalizability study (G-study) against human labels, invariance and DIF tests, and linking onto a common scale. In a known-truth simulation with 500 replications, pooling platform scores at face value reported the wrong direction of daily sentiment on 13.8 of 60 days on average, against 0.8 days for an adjusted index. The paper came from my MSc dissertation (255,446 comments on UK economic policy across X, Reddit, YouTube and Quora), where I found that my classifier's 88 per cent accuracy only measured agreement with the lexicon that produced its training labels. Two gaps remain: the protocol has not been run on real data, and it has not been extended to languages, code-mixing and changing LLM annotators.

3. Research questions

1. **Measurement.** How much of the variance in LLM-assigned sentiment, stance and narrative labels in South and Southeast Asian political talk comes from the platform, the language or code-mixed register, and the choice of annotator model and prompt, rather than from the content? Which constructs are measurement-equivalent?
2. **Audiences.** Are platform differences in measurement really differences in audience composition? Do cross-platform audience-overlap networks explain where labels stop being comparable?
3. **Consequences.** When opinion and polarisation indicators are corrected and linked, do substantive conclusions change, for example about the direction of public reaction or the spread of a narrative around an election?

4. Data and methods

Study 1 (simulation, year 1). Extend the existing simulation so that platform, language, annotator model, prompt and time are all facets, and estimate how many human-labelled posts, and which anchor events, are needed to detect a difference large enough to reverse a conclusion. This needs no new data and sets the design for the empirical work.

Study 2 (empirical, years 2 to 3). Collect public posts around political events fixed in advance (for example, elections and major policy announcements in Pakistan, India and Singapore) from several platforms, including X, Facebook pages, YouTube comments and public Telegram or WhatsApp channels where access is ethical and permitted, through official APIs and permitted research access under NUS ethics review. Label every post with several LLMs and fine-tuned models, and draw a stratified sample per platform and language (English, Urdu, Hindi, and code-mixed Roman Urdu, Hinglish and Singlish) for blind labelling by trained native-speaker coders, with a second coder on a subset. Run a G-study to split variance into platform, language, model, prompt and time; test invariance and DIF for each construct; and correct downstream estimates with the human sample (Egami et al., 2023). Linear concept erasure (Fan et al., 2025) will test whether removing the platform signal also removes real differences.

Study 3 (substantive, years 3 to 4). Build co-exposure and page-affinity networks across platforms, detect communities, and model whether label shifts follow community membership (RQ2). Then re-run a polarisation or narrative-spread analysis around an election window with corrected and uncorrected measures, using interrupted time series, and report when the conclusion changes (RQ3).

5. Why NUS CNM, and feasibility

The programme is described as highly suitable for computational communication and political communication, and this project sits across the Politics, Participation and Governance cluster (how public opinion is read from platforms) and the Technology, Design and Social Futures cluster (responsible design of AI-based measurement). Dr Subhayan Mukerjee's work on audiences across platforms, network methods and LLMs in online news is the closest fit for the project as a whole, and Dr Zening Duan's work on validating computational and LLM-based measures of content fits RQ1. Work by Associate Professors Taberez Neyazi, Kokil Jaidka and Ozan Kuru on South and Southeast Asian publics, multilingual text and public opinion measurement would inform the design, and the Centre for Trusted Internet and Community and SICSS-Singapore offer a natural community.

On feasibility: I have collected and labelled public web and social data for nearly eight years, at Dubizzle Labs (data in English, Arabic, Urdu and Hindi across nine countries), at M+C Saatchi Fluency (LLM tagging with prompt-version tracking for global brands) and as co-founder and CTO of Artemis AI, where I lead a system that builds page-affinity networks with community detection and where our own evaluation showed that agreement between models is not accuracy. I speak Urdu and Hindi, which helps with gold-set design. The simulation code is public, and the statistical methods are established; what is new is applying them with platforms, languages and models as the groups.

6. Expected contribution

A tested protocol and open toolkit for checking when machine-labelled public opinion can be compared; native-speaker gold sets for code-mixed South Asian political text; evidence on how much platform, language and model choice distort findings; and practical guidance for researchers, newsrooms and public bodies that rely on social listening. A first paper is planned for *Computational Communication Research* or *Social Science Computer Review*.

Selected references

Chen, K., Duan, Z. and Kim, S. J. (2024). Uncovering gender stereotypes in controversial science discourse: evidence from computational text and visual analyses across digital platforms. Journal of Computer-Mediated Communication 29(1), zmad052.

Duan, Z. et al. (2025). Constructing vec-tionaries to extract message features from texts: a case study of moral content. Political Analysis.

Egami, N., Hinck, M., Stewart, B. M. and Wei, H. (2023). Using imperfect surrogates for downstream inference: design-based supervised learning for social science applications of large language models. NeurIPS.

Fan, Y. et al. (2025). The medium is not the message: deconfounding document embeddings via linear concept erasure. EMNLP.

Gilardi, F., Alizadeh, M. and Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. PNAS 120(30).

Ikram, A. and Haig, E. (2026). Is the platform part of the measurement? A protocol and simulation study for cross-platform equivalence of machine-labelled policy sentiment. arXiv:2610.16757.

Jacobs, A. Z. and Wallach, H. (2021). Measurement and fairness. ACM FAccT.

Jaidka, K., Guntuku, S. C. and Ungar, L. H. (2018). Facebook versus Twitter: differences in self-disclosure and trait prediction. ICWSM.

Mukerjee, S., Yang, T. and Peng, Y. (2023). Metrics in action: how social media metrics shape news production on Facebook. Journal of Communication 73(3).

Neyazi, T. A., Tan, K. E. and Kuru, O. (2026). The interplay of conservatism and religiosity on misinformation susceptibility: evidence from the 2022 Malaysian general election. International Journal of Public Opinion Research 38(1).

Ouyang, R., Mukerjee, S. and Yang, T. (2026). How social media metrics shape news production: a replication study. The International Journal of Press/Politics.